Intro to Machine Learning
Practice Problems 2: Ridge Regression, the Lasso and the Elastic Net

Practice problems, not turned in

Directions: These practice problems go with Week 2 (ridge regression, the lasso and the elastic net). They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Each problem is adapted from the exercise cited next to it; see the book for the original wording.

Throughout, n is the number of observations, p the number of predictors, and the intercept β_0 is never penalized. Unless a problem says otherwise, ridge and the lasso use ISL's scaling:

$$\text{ridge: } \mathrm{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2, \qquad \text{lasso: } \mathrm{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|, \qquad \mathrm{RSS} = \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j \Big)^2.$$

1. **More or less flexible than least squares?** (ISL 6.6, Exercise 2(a)–(b); ISL 6.2.) For each method below, choose the one correct statement and justify it.

   (a) The lasso, compared with least squares, is

   i. more flexible, so it predicts better when its increase in bias is smaller than its decrease in variance;

   ii. more flexible, so it predicts better when its increase in variance is smaller than its decrease in bias;

   iii. less flexible, so it predicts better when its increase in bias is smaller than its decrease in variance;

   iv. less flexible, so it predicts better when its increase in variance is smaller than its decrease in bias.

   (b) The same question for ridge regression compared with least squares.

2. **The lasso as the budget grows.** (ISL 6.6, Exercise 3; ISL 6.2.2.) Fit the lasso in its constrained form,

$$\text{minimize RSS} \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \le s,$$

   and increase s from 0. For each quantity below, say whether it (i) increases and then decreases (an inverted U), (ii) decreases and then increases (a U), (iii) steadily increases, (iv) steadily decreases, or (v) stays constant. Justify each answer.

   (a) training RSS (b) test RSS (c) variance (d) squared bias (e) irreducible error

3. **Ridge as the penalty grows.** (ISL 6.6, Exercise 4; ISL 6.2.1.) Now fit ridge regression in its penalized form, $\text{RSS} + \lambda \sum_j \beta_j^2$, and increase λ from 0. Answer the same five parts as Problem 2, using the same options i–v.